

Exam on Dynamical Systems
July 8, 2025

1. (a) (5p) Let $u: [0, \infty) \rightarrow \mathbb{R}$ be a C^1 -function such that both limits $\lim_{t \rightarrow \infty} u(t)$ and $\lim_{t \rightarrow \infty} \dot{u}(t)$ exists and are finite. Prove that $\lim_{t \rightarrow \infty} \dot{u}(t) = 0$.
- (b) (5p) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a C^1 function and $u: [0, \infty) \rightarrow \mathbb{R}$ be a solution of the differential equation $\dot{x} = f(x)$. Assume that there exists $\lim_{t \rightarrow \infty} u(t) = 2$. Prove that 2 is an equilibrium point of $\dot{x} = f(x)$.
- (c) (5p) Let $f(x) = 4 - x^2$. Find the solution of the IVP $\dot{x} = f'(2)x$, $x(0) = 1$ and its limit as $t \rightarrow \infty$.
- (d) (5p) We consider the scalar dynamical system $\dot{x} = 4 - x^2$. Find the equilibria and study their stability using the linearization method. Represent the phase portrait.
- (e) (5p) Let $u: \mathbb{R} \rightarrow \mathbb{R}$ be the solution of the IVP

$$\dot{x} = 4 - x^2, \quad x(0) = 0.$$

Reading the phase portrait of $\dot{x} = 4 - x^2$, find the image of u , its intervals of monotonicity and its limits at $\pm\infty$. Using these informations, represent the graph of u .

2. Let $\omega > 0$ be a fixed parameter. The second order linear equation $\theta'' + \omega^2\theta = 0$ describes the *small* oscillations of a simple idealized pendulum.

- (a) (10p) Find the solution of the IVP

$$\theta'' + \omega^2\theta = 0, \quad \theta(0) = \frac{\pi}{6}, \quad \theta'(0) = 0.$$

Describe the motion of the pendulum when $\theta(t)$, the solution found, is the angle (measured in radians) between the rod and the vertical line. What is the period of the motion (i.e. after how much time the pendulum will come back to the initial position)?

- (b) (10p) Represent the phase portrait of the associated planar dynamical system (finding a first integral). Specify its type and stability. Specify the shape of the orbits.

3. We consider the map

$$T: \mathbb{R} \rightarrow \mathbb{R}, \quad T(x) = 1 - |2x - 1|.$$

- a) (8p) Represent the graph of T . Find the fixed points of T .
- b) (4p) Let $n \in \mathbb{N}$, $n \geq 2$. Find the orbit of the initial state $\frac{3}{2^n}$.
- c) (3p) Find the 2-periodic points of T .

Seminar test. Find the solution of the IVP

- (a) (5p) $x' + 2tx = t$, $x(0) = 0$.
- (b) (5p) $x' - tx = 2$, $x(0) = 0$.

Exam on Dynamical Systems
June 26, 2025

- 1.** Let $\lambda \in \mathbb{R}$ be an eigenvalue and $v \in \mathbb{R}^2$ be a corresponding eigenvector of a matrix $B \in \mathcal{M}_2(\mathbb{R})$.
(a) (6p) Find $\varphi: \mathbb{R} \rightarrow \mathbb{R}$ such that $\psi(t) = \varphi(t)v$ is the solution of the IVP

$$X' = BX, \quad X(0) = v.$$

Justify that

$$e^{tB}v = e^{t\lambda}v, \quad t \in \mathbb{R}.$$

- (b) (4p) Find $\varphi: \mathbb{N} \rightarrow \mathbb{R}$ such that $\psi_k = \varphi_k v$ is the solution of the IVP

$$X_{k+1} = BX_k, \quad X_0 = v.$$

Justify that

$$B^k v = \lambda^k v, \quad k \in \mathbb{N}.$$

- (c) (6p) Find $\varphi: \mathbb{R} \rightarrow \mathbb{R}$ such that $u(t, x) = \varphi(t) \sin(2x)$ verifies (the heat equation)

$$\frac{\partial u}{\partial t}(t, x) - 3 \frac{\partial^2 u}{\partial x^2}(t, x) = 0, \quad (t, x) \in \mathbb{R}^2.$$

- (d) (8p) Find $\varphi: \mathbb{R} \rightarrow \mathbb{R}$ such that $u(t, x) = \varphi(t) \sin(x)$ verifies (the wave equation)

$$\frac{\partial^2 u}{\partial t^2}(t, x) - 3 \frac{\partial^2 u}{\partial x^2}(t, x) = 0, \quad (t, x) \in \mathbb{R}^2.$$

- 2.** Consider the linear planar system $\dot{X} = AX$ with $A = \begin{pmatrix} 2 & -3 \\ 1 & -2 \end{pmatrix}$. Let $v_1, v_2 \in \mathbb{R}^2$ be two linearly independent eigenvectors of A .

- (a) (4p) Specify its type and stability.

- (b) (5p) Find $\varphi(t, \eta)$, its flow.

- (c) (8p) Represent the orbits corresponding to the initial values $v_1, -v_1, v_2$ and, respectively, $-v_2$.

- (d) (8p) Find $a, b \in \mathbb{R}$ such that $H(x, y) = ax^2 + bxy + 3y^2$ is a first integral. Find $H(\varphi(t, v_1))$, $H(\varphi(t, v_2))$ and represent the level curve $H(x, y) = 0$.

- 3.** (11p) Find a range of values for h such that the attractor equilibrium point of $x' = x^2 + 5x + 6$ is also an attractor fixed point of the discrete dynamical system associated to the Euler's numerical formula with stepsize $h > 0$ for the given differential equation.

Exam on Dynamical Systems
June 26, 2025

1. Let $\lambda \in \mathbb{R}$ be an eigenvalue and $u \in \mathbb{R}^2$ be a corresponding eigenvector of a matrix $A \in \mathcal{M}_2(\mathbb{R})$.

(a) (6p) Find $b: \mathbb{R} \rightarrow \mathbb{R}$ such that $\varphi(t) = b(t)u$ is the solution of the IVP

$$X' = AX, \quad X(0) = u.$$

Justify that

$$e^{tA}u = e^{t\lambda}u, \quad t \in \mathbb{R}.$$

(b) (4p) Find $b: \mathbb{N} \rightarrow \mathbb{R}$ such that $\varphi_k = b_k u$ is the solution of the IVP

$$X_{k+1} = AX_k, \quad X_0 = u.$$

Justify that

$$A^k u = \lambda^k u, \quad k \in \mathbb{N}.$$

(c) (6p) Find $b: \mathbb{R} \rightarrow \mathbb{R}$ such that $v(t, x) = b(t) \sin(3x)$ verifies (the heat equation)

$$\frac{\partial v}{\partial t}(t, x) - \frac{\partial^2 v}{\partial x^2}(t, x) = 0, \quad (t, x) \in \mathbb{R}^2.$$

(d) (8p) Find $b: \mathbb{R} \rightarrow \mathbb{R}$ such that $v(t, x) = b(t) \sin(3x)$ verifies (the wave equation)

$$\frac{\partial^2 v}{\partial t^2}(t, x) - \frac{\partial^2 v}{\partial x^2}(t, x) = 0, \quad (t, x) \in \mathbb{R}^2.$$

2. Consider the linear planar system $\dot{X} = AX$ with $A = \begin{pmatrix} -1 & -3 \\ -1 & 1 \end{pmatrix}$. Let $u_1, u_2 \in \mathbb{R}^2$ be two linearly independent eigenvectors of A .

(a) (4p) Specify its type and stability.

(b) (5p) Find $\varphi(t, \eta)$, its flow.

(c) (8p) Represent the orbits corresponding to the initial values $u_1, -u_1, u_2$ and, respectively, $-u_2$.

(d) (8p) Find $b, c \in \mathbb{R}$ such that $H(x, y) = x^2 + bxy + cy^2$ is a first integral. Find $H(\varphi(t, u_1))$, $H(\varphi(t, u_2))$ and represent the level curve $H(x, y) = 0$.

3. (11p) Find a range of values for h such that the attractor equilibrium point of $x' = -x^2 + 5x - 6$ is also an attractor fixed point of the discrete dynamical system associated to the Euler's numerical formula with stepsize $h > 0$ for the given differential equation.

Exam on Dynamical Systems
June 23, 2025

1. Let $\varphi: \mathbb{R} \rightarrow \mathbb{R}$, $\varphi(t) = e^{-t} \cos(2t)$.

(a) (4p) Represent the graph of φ . Describe the motion of a simple planar pendulum for which $\varphi(t)$ is the angle (measured in radians) between the rod and the vertical at the time t (measured in seconds).

(b) (8p) Find the second order linear homogeneous differential equation with constant coefficients for which φ is a solution. Justify that the general solution of the DE found is described by the formula

$$Ae^{-t} \cos(2t - \theta), \quad \text{where } A \geq 0, \theta \in [0, 2\pi) \quad \text{are arbitrary constants.}$$

(c) (8p) We consider the linear planar differential system

$$\dot{x} = -x + 2y, \quad \dot{y} = -2x - y.$$

Find its general solution. Represent the orbit corresponding to the initial value $\eta = (1, 0) \in \mathbb{R}^2$.

2. We consider the map $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x - \frac{1}{4}(x^2 - 2)$.

(a) (5p) Prove that there is no $\eta \in \mathbb{R}$ such that $\lim_{k \rightarrow \infty} f^k(\eta) = 2$.

(b) (5p) Justify that there exists $\eta \in \mathbb{R} \setminus \{\sqrt{2}\}$ such that $\lim_{k \rightarrow \infty} f^k(\eta) = \sqrt{2}$.

(c) (5p) Represent the graph of f .

(d) (5p) Using the cobweb diagram, depict $f^k(0)$ for $k \in \{1, 2, 3\}$. Describe the long-term behavior of the sequence $(f^k(\eta))_{k \geq 0}$.

3. How many solutions has the following problem:

(a) (5p) $x'' + tx' + x = 0$, $x(0) = 1$, $x'(0) = -2$.

(b) (5p) $x'' + tx' + x = 0$, $x(0) = 1$.

(c) (2p) $x'' + tx' + x = 0$, $x(0) = 1$, $x'(0) = -2$, $x''(0) = -1$.

(d) (2p) $x'' + tx' + x = 0$, $x(0) = 1$, $x'(0) = -2$, $x''(0) = -2$.

(e) (2p) $x'' + tx' + x = 0$, $x(0) = 1$, $x''(0) = -2$.

(f) (2p) $x'' + tx' + x = 0$, $x(0) = 1$, $x''(0) = -1$.

(g) (2p) $x'' + tx' + x = 0$, $x''(0) = -2$.

Exam on Dynamical Systems
June 16, 2025

1. (a) (5p) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a C^1 function and $u: [0, \infty) \rightarrow \mathbb{R}$ be a solution of the differential equation $\dot{x} = f(x)$. Assume that there exists $\lim_{t \rightarrow \infty} u(t) = 2$. Prove that 2 is an equilibrium point of $\dot{x} = f(x)$.
- (b) (5p) Let $f(x) = 2x - x^2$. Find the solution of the IVP $\dot{x} = f'(2)x$, $x(0) = 1$ and its limit as $t \rightarrow \infty$.
- (c) (5p) We consider the scalar dynamical system $\dot{x} = 2x - x^2$. Find the equilibria and study their stability using the linearization method. Represent the phase portrait.
- (d) (5p) Let $u: \mathbb{R} \rightarrow \mathbb{R}$ be the solution of the IVP

$$\dot{x} = 2x - x^2, \quad x(0) = 1.$$

Reading the phase portrait, find the image of u , its intervals of monotonicity and its limits at $\pm\infty$. Using these, represent the graph of u .

2. (a) (5p) We consider the following IVP, where the unknown is a complex function $z: \mathbb{R} \rightarrow \mathbb{C}$, $t \mapsto z(t)$

$$\dot{z} = (-2 + 2i)z, \quad z(0) = 1.$$

Find the solution, its value in $\frac{\pi}{2}$, and represent its image in the complex plane.

- (b) (5p) Write the real planar system equivalent to the complex equation $\dot{z} = (-2 + 2i)z$ (use the notation $z = x + iy$). Specify its type and stability.

- (c) (10p) We consider the following IVP, where the unknown is a complex sequence $z: \mathbb{N} \rightarrow \mathbb{C}$, $k \mapsto z_k$

$$z_{k+1} = (-2 + 2i)z_k, \quad z_0 = 1.$$

Find the solution and represent its image in the complex plane.

3. Let $g_\lambda: [0, 1] \rightarrow \mathbb{R}$, $g_\lambda(x) = \lambda x(1 - x)$. Let $\eta_1 = \sin^2 \frac{\pi}{5}$, $\eta_2 = \sin^2 \frac{\pi}{9}$.

- (a) (10p) Prove that g_3 has no 2-cycle.

- (b) (8p) Prove that γ_{η_1} is a 2-cycle of g_4 and γ_{η_2} is a 3-cycle of g_4 .

- (c) (2p) Represent the graphs of g_3 and g_4 .

Exam on Dynamical Systems
June 16, 2025

- 1.** (a) (5p) Let $g: \mathbb{R} \rightarrow \mathbb{R}$ be a C^1 function and $\varphi: [0, \infty) \rightarrow \mathbb{R}$ be a solution of the differential equation $\dot{x} = g(x)$. Assume that there exists $\lim_{t \rightarrow -\infty} \varphi(t) = 3$. Prove that 3 is an equilibrium point of $\dot{x} = g(x)$.
- (b) (5p) Let $g(x) = x^2 - 3x$. Find the solution of the IVP $\dot{x} = g'(3)x$, $x(0) = 2$ and its limit as $t \rightarrow -\infty$.
- (c) (5p) We consider the scalar dynamical system $\dot{x} = x^2 - 3x$. Find the equilibria and study their stability using the linearization method. Represent the phase portrait.
- (d) (5p) Let $\varphi: \mathbb{R} \rightarrow \mathbb{R}$ be the solution of the IVP $\dot{x} = x^2 - 3x$, $x(0) = 2$. Reading the phase portrait, find the image of φ , its intervals of monotonicity and its limits at $\pm\infty$. Using these, represent the graph of φ .

- 2.** (a) (5p) We consider the following IVP, where the unknown is a complex function $z: \mathbb{R} \rightarrow \mathbb{C}$, $t \mapsto z(t)$

$$\dot{z} = \left(\frac{1}{2} + \frac{i}{2}\right)z, \quad z(0) = 1.$$

Find the solution, its value in 2π , and represent its image in the complex plane.

- (b) (5p) Write the real planar system equivalent to the complex equation $\dot{z} = \left(\frac{1}{2} + \frac{i}{2}\right)z$ (use the notation $z = x + iy$). Specify its type and stability.
- (c) (10p) We consider the following IVP, where the unknown is a complex sequence $z: \mathbb{N} \rightarrow \mathbb{C}$, $k \mapsto z_k$

$$z_{k+1} = \left(\frac{1}{2} + \frac{i}{2}\right)z_k, \quad z_0 = 1.$$

Find the solution and represent its image in the complex plane.

- 3.** Let $f_\lambda: [0, 1] \rightarrow \mathbb{R}$, $f_\lambda(x) = \lambda x(1 - x)$. Let $\eta_1 = \sin^2 \frac{\pi}{5}$, $\eta_2 = \sin^2 \frac{\pi}{9}$.

- (a) (10p) Prove that f_3 has no 2-cycle.
- (b) (8p) Prove that γ_{η_1} is a 2-cycle of f_4 and γ_{η_2} is a 3-cycle of f_4 .
- (c) (2p) Represent the graphs of f_3 and f_4 .

Exam on Dynamical Systems
June 10, 2025

1. Find the solution of the IVP

- (a) (5p) $x' = -2x$, $x(0) = 3$;
- (b) (10p) $x' = 4x - y$, $y' = x + 4y$, $x(0) = 0$, $y(0) = 3$;
- (c) (5p) $x_{k+2} - 4x_{k+1} + 4x_k = 0$, $x_0 = 0$, $x_1 = 3$.

2. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = 3 + 0.7 \sin x$, $x \in \mathbb{R}$. Prove that

- (a) (5p) $|f(x) - f(y)| \leq 0.7|x - y|$ for all $x, y \in \mathbb{R}$;
- (b) (5p) f has a unique fixed point, denoted by η^* ;
- (b) (10p) η^* is a global attractor fixed point of the discrete dynamical system $x_{k+1} = f(x_k)$.

3. Consider the following planar system

$$\dot{x} = -2y(x^2 + 2y^2), \quad \dot{y} = x(x^2 + 2y^2).$$

- (a) (5p) Decide whether the equilibrium point $(0, 0)$ is hyperbolic or not.
- (b) (5p) Verify that $\varphi(t, \sqrt{2}, 0) = (\sqrt{2} \cos(t\sqrt{8}), \sin(t\sqrt{8}))$. Represent the orbit corresponding to the initial point $(\sqrt{2}, 0)$.
- (c) (5p) Find a global first integral.
- (d) (3p) Represent its phase portrait.
- (e) (2p) Describe the long-term behavior of the solutions of this system.

Exam on Dynamical Systems
June 10, 2025

1. Find the solution of the IVP

- (a) (5p) $x_{k+1} = 2x_k, x(0) = 3$;
- (b) (5p) $x' + tx = 1, x(0) = 0$;
- (c) (10p) $x_{k+2} - x_{k+1} + x_k = 0, x_0 = 0, x_1 = 3$.

2. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = 4 + 0.5 \cos x, x \in \mathbb{R}$. Prove that

- (a) (5p) $|f(x) - f(y)| \leq 0.5|x - y|$ for all $x, y \in \mathbb{R}$;
- (b) (5p) f has a unique fixed point, denoted by η^* ;
- (b) (10p) η^* is a global attractor fixed point of the discrete dynamical system $x_{k+1} = f(x_k)$.

3. Consider the following planar system

$$\dot{x} = y(3x^2 + y^2), \quad \dot{y} = -3x(3x^2 + 2y^2).$$

- (a) (5p) Decide whether the equilibrium point $(0, 0)$ is hyperbolic or not.
- (b) (5p) Verify that $\varphi(t, 0, \sqrt{3}) = (\frac{1}{\sqrt{3}} \sin(t\sqrt{3}), \cos(t\sqrt{3}))$. Represent the orbit corresponding to the initial point $(0, \sqrt{3})$.
- (c) (5p) Find a global first integral.
- (d) (3p) Represent its phase portrait.
- (e) (2p) Describe the long-term behavior of the solutions of this system.

Exam on Dynamical Systems
July 13, 2024

Name _____
Group _____

1. (15p) The error function is defined by $\operatorname{erf}(t) = \frac{2}{\sqrt{\pi}} \int_0^t e^{-s^2} ds$ for all $t \in \mathbb{R}$. Express the solution of the IVP $x' - 2tx = 1$, $x(1) = 0$ in terms of the error function.

2. We consider the nonlinear planar system

$$\dot{x} = -y - x(x^2 + y^2), \quad \dot{y} = x - y(x^2 + y^2).$$

(a) (10p) Justify that $(0, 0)$ is an equilibrium point which is not hyperbolic.

(b) (10p) Represent the phase portrait (using polar coordinates). Reading the phase portrait specify the stability of the equilibrium point $(0, 0)$.

(c) (10p) Consider now the linearized system around the equilibrium point $(0, 0)$. Find its general solution and represent its phase portrait. Specify its stability.

3. (15p) Let $\eta > 0$ be sufficiently small and $f : \mathbb{R} \rightarrow \mathbb{R}$ be the function defined by $f(x) = x^2 - 3$. We consider a sequence $(x_k)_{k \in \mathbb{N}}$ such that $x_0 = \eta + \sqrt{3}$ and $x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$ for all $k \in \mathbb{N}$. Using the linearization method, justify that this sequence is convergent.

What can you say about the sequence $(x_k)_{k \in \mathbb{N}}$ defined by $x_0 = \eta$ and $x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$ for all $k \in \mathbb{N}$ constructed with $f(x) = x^{1/3}$?

Seminar Test. (10p) Let $k > 0$ be a fixed parameter. Find the flow $\varphi(t, \eta)$ of the scalar dynamical system $x' = -k(x - 13)$. Compute $\lim_{t \rightarrow \infty} \varphi(t, \eta)$ for any $\eta \in \mathbb{R}$. Represent the phase portrait.